

Pre-Flight Checklist

Mechanical & Structural Inspection

Inspection Task	Technical Requirement / Specification	Status
Airframe Integrity	Verify F450 frame for cracks; confirm structural symmetry.	☐ ▾
Vibration Damping	Inspect anti-vibration pads for the flight controller.	☐ ▾
Propulsion Hardware	Check 4x A2212 1000KV motors for friction or wobble.	☐ ▾
Propeller Mounting	Verify 1045 props (2 CW, 2 CCW) are securely fastened and free of nicks.	☐ ▾

Power Management System

Inspection Task	Technical Requirement / Specification	Status
Battery Voltage	Verify the given battery is above > 75 percent of charge.	☐ ▾
Safety Thresholds	Confirm low-voltage failsafe is set to 40 percent of the battery.	☐ ▾
Physical Connections	Ensure XT60 connectors are secure with no abrupt disconnections.	☐ ▾
PDB Status	Inspect PDB for shorts and loose connections.	☐ ▾

Avionics & Flight Controller

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Inspection Task	Technical Requirement / Specification	Status
Controller Init	Pixhawk 2.4.8 initialization; verify status LED sequence.	☐ ▾
GPS Acquisition	Confirm GPS lock with accuracy within 2-3 meters.	☐ ▾
Sensor Fusion	Check EKF (Extended Kalman Filter) health via telemetry.	☐ ▾
Compass	Verify compass calibration, with drift of +/- 3 degrees max to max. If fast drifting abort flight. Same goes For Roll/Pitch from IMUs	☐ ▾

Ground Control Station & Pre-Launch

Inspection Task	Technical Requirement / Specification	Status
Control Link	Verify 2.4 GHz RC link.	☐ ▾
GCS Sync	Establish Mission Planner connection; verify MavLink data stream.	☐ ▾
Geo-Fencing	Define and upload virtual boundaries to the flight controller.	☐ ▾
Failsafe Audit	Confirm Return-to-Home (RTH) triggers on signal loss or critical voltage.	☐ ▾
Pre-Launch Mode	Always Arm in /Loiter.	☐ ▾
Status messages check	If any IMU/Compass warnings are given abort flight.	☐ ▾